

Lead Conversion Analysis: Answers to Assignment Questions

May 29, 2025

Introduction

This document provides answers to the subjective questions based on a logistic regression model built to predict lead conversion for X Education. The model was trained on the `Leads.csv` dataset after cleaning and feature engineering, using Recursive Feature Elimination (RFE) to select relevant features. The model demonstrates strong performance with the following metrics:

- **Accuracy:** 0.9345
- **ROC AUC Score:** 0.9712
- **Precision (Class 1):** 0.9405
- **Recall (Class 1):** 0.8813

Question 1: Top 3 Variables Contributing Most to Conversion Probability

The top three variables with the largest positive, statistically significant coefficients (p-value < 0.05) in the logistic regression model are:

1. **Tags_Will revert after reading the email** (Coefficient: 6.4769, $p < 0.001$): Leads with this tag are highly likely to convert, indicating strong interest post-initial contact.
2. **Tags_Closed by Horizzon** (Coefficient: 9.5605, $p < 0.001$): This tag shows a strong positive association with conversion, possibly reflecting a successful segment or process.
3. **Lead Source_Welingak Website** (Coefficient: 4.2551, $p < 0.001$): Leads from this source have a significantly higher conversion probability.

Note: Tags_Lost to EINS (Coef: 8.1986, $p < 0.001$) is also significant but ranked below Closed by Horizzon due to its slightly lower coefficient. Tags with high coefficients but non-significant p-values (e.g., Tags_Lateral student) were excluded due to issues like multicollinearity or low frequency.

Question 2: Top 3 Categorical/Dummy Variables to Focus On

To increase conversion probability, focus on the following categorical/dummy variables:

1. **Tags__Will revert after reading the email:** Prioritize nurturing these leads and replicate the processes leading to this tag.
2. **Tags__Closed by Horizzon:** Investigate the processes or segments associated with this tag to leverage its high conversion rate.
3. **Lead Source__Welingak Website:** Intensify marketing through this channel and analyze its user experience for broader application.

Question 3: Strategy for Aggressive Conversion (Intern Period)

Goal: Maximize true positives (high recall) during a 2-month intern period.

1. **Use Predicted Probabilities:** Apply the logistic regression model to predict conversion probabilities.
2. **Lower Probability Threshold:** Reduce the classification threshold to 0.30–0.35 to expand the pool of leads classified as potential converters.
3. **Identify Optimal Threshold:** Adjust based on intern call capacity and precision-recall trade-off, ensuring high recall with acceptable precision.
4. **Prioritize within Selected Pool:** Rank leads by predicted probability, presence of key tags (e.g., Will revert, Closed by Horizzon), and high Total Time Spent on Website.

Outcome: Increases true positives, though with more false positives, suitable for aggressive conversion with intern resources.

Question 4: Strategy for Minimizing Useless Calls (Target Reached Period)

Goal: Minimize false positives (high precision) when sales targets are met.

1. **Use Predicted Probabilities:** Use model predictions for lead conversion probabilities.
2. **Raise Probability Threshold:** Set a high threshold (0.70–0.75) to classify only high-confidence leads as converters.
3. **Identify Optimal Threshold:** Select a threshold yielding high precision (>95%) via the precision-recall curve.
4. **Focus on High-Confidence Leads:** Contact only leads exceeding this threshold.

Outcome: Minimizes wasted calls by focusing on leads most likely to convert, accepting lower recall for efficiency.

Supporting Files

- `leads_cleaned.csv`: Cleaned and engineered dataset.
- `logistic_model_results.pkl`: Saved statsmodels results.
- `selected_features.txt`: RFE-selected features.
- `test_predictions.csv`: Actual vs. predicted test set values.
- `scaler.joblib`: Saved StandardScaler object.